

Evaluating Short-Term Forecasting of Multiple Time Series in IoT Environments

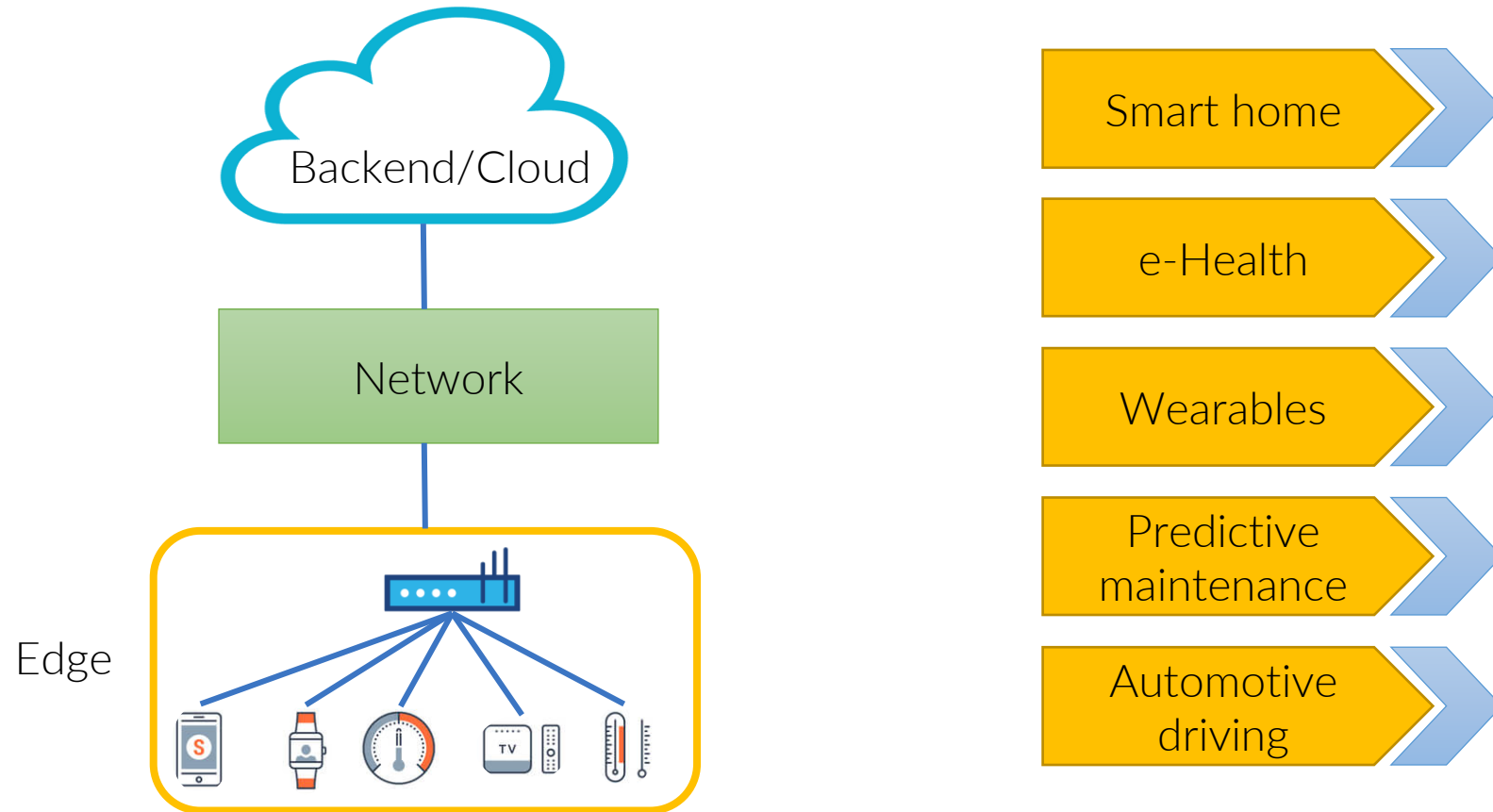
Christos Tzagkarakis, Pavlos Charalampidis, Stylianos Roubakis,
Alexandros Fragkiadakis and Sotiris Ioannidis



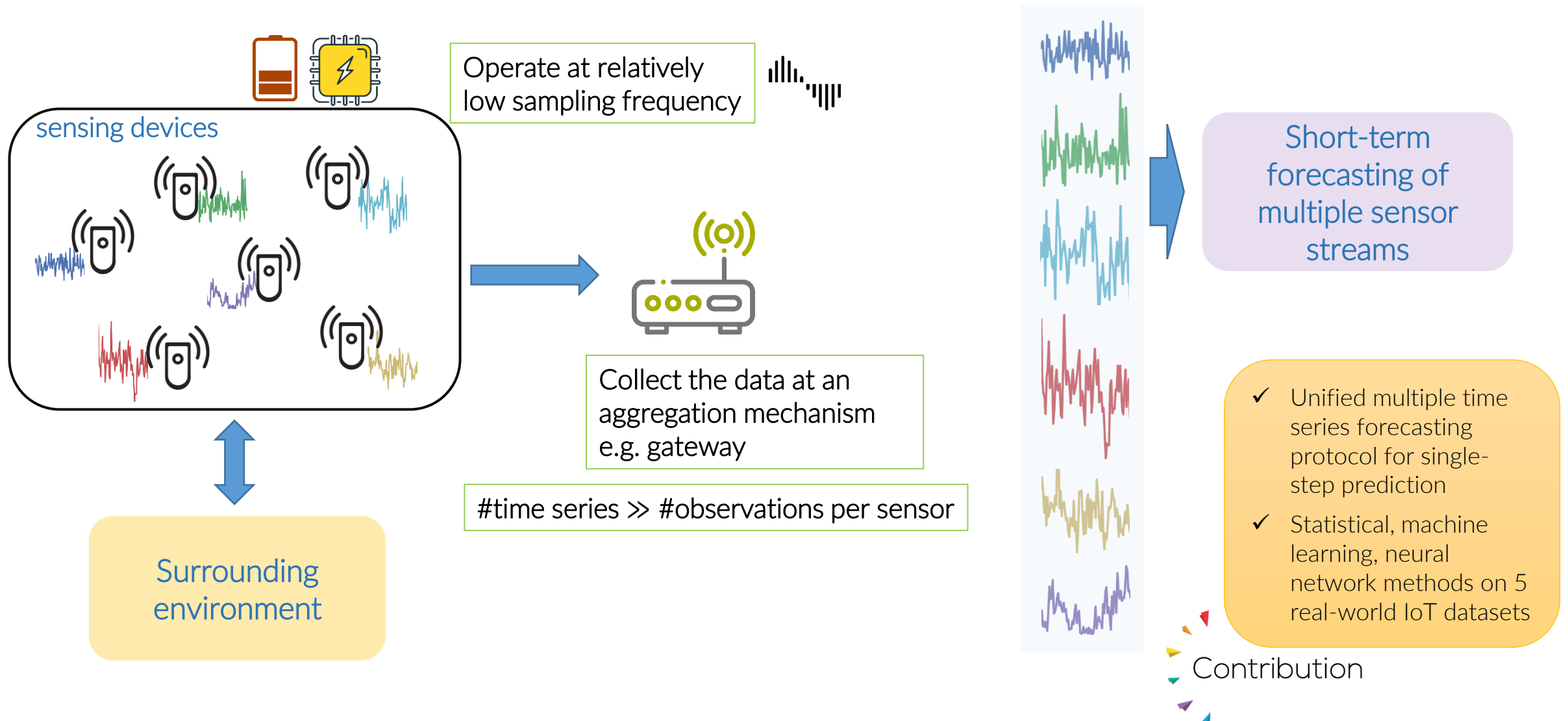
- Introduction
- Problem formulation
- Forecasting protocol
- Datasets
- Experimental results



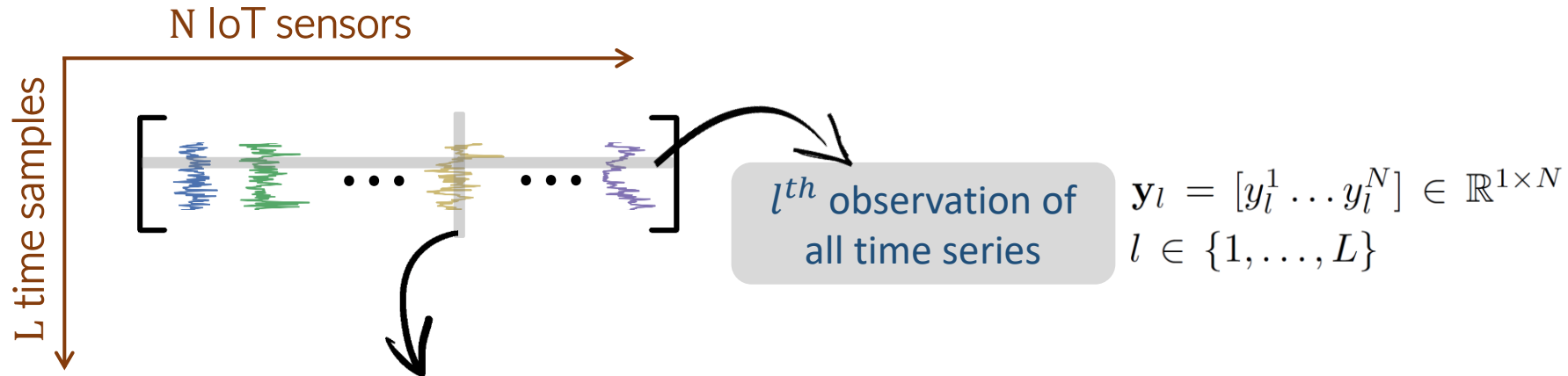
 Internet-of-Things (IoT) has advanced the concept of Internet networking



Introduction



Problem formulation



time series captured by the n^{th} sensor

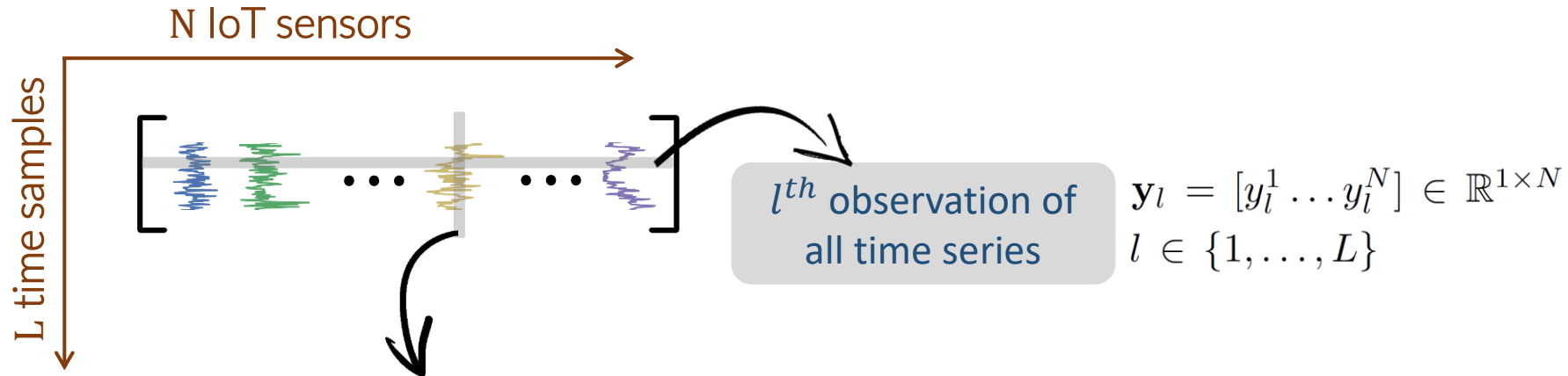
$\mathbf{y}^n = [y_1^n \dots y_L^n]^T \in \mathbb{R}^{N \times 1}$
 $n \in \{1, \dots, N\}$

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_l \\ \vdots \\ \mathbf{y}_L \end{bmatrix} = \begin{bmatrix} y_1^1 & y_1^2 & \dots & y_1^N \\ \vdots & \vdots & & \vdots \\ y_l^1 & y_l^2 & \dots & y_l^N \\ \vdots & \vdots & & \vdots \\ y_L^1 & y_L^2 & \dots & y_L^N \end{bmatrix} \in \mathbb{R}^{L \times N}$$

forecasting horizon h

$$\mathbf{Y}^{(f)} = \begin{bmatrix} \mathbf{y}_{L+1} \\ \vdots \\ \mathbf{y}_{L+h} \end{bmatrix} = \begin{bmatrix} y_{L+1}^1 & y_{L+1}^2 & \dots & y_{L+1}^N \\ \vdots & \vdots & & \vdots \\ y_{L+h}^1 & y_{L+h}^2 & \dots & y_{L+h}^N \end{bmatrix} \in \mathbb{R}^{h \times N}$$

Problem formulation



$$\mathbf{y}_l = [y_l^1 \dots y_l^N] \in \mathbb{R}^{1 \times N}$$
$$l \in \{1, \dots, L\}$$

$$\mathbf{y}^n = [y_1^n \dots y_L^n]^T \in \mathbb{R}^{N \times 1}$$
$$n \in \{1, \dots, N\}$$

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_l \\ \vdots \\ \mathbf{y}_L \end{bmatrix} = \begin{bmatrix} y_1^1 & y_1^2 & \dots & y_1^N \\ \vdots & \vdots & & \vdots \\ y_l^1 & y_l^2 & \dots & y_l^N \\ \vdots & \vdots & & \vdots \\ y_L^1 & y_L^2 & \dots & y_L^N \end{bmatrix} \in \mathbb{R}^{L \times N}$$

forecasting horizon $h = 1$

$$\mathbf{y}_{L+1} = [y_{L+1}^1 y_{L+1}^2 \dots y_{L+1}^N] \in \mathbb{R}^{1 \times N}$$

Problem formulation



Compute a **forecasting model** $\mathcal{M}$ based on historical observations of the multiple time series $\mathbf{Y}$ to predict the future values y_{L+1}

model parameters

$$\hat{\mathbf{y}}_{L+1} = [\hat{y}_{L+1}^1 \hat{y}_{L+1}^2 \cdots \hat{y}_{L+1}^N] \in \mathbb{R}^{1 \times N}$$

$$\mathcal{M}(\mathbf{Y}, \boldsymbol{\theta})$$

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_l \\ \vdots \\ \mathbf{y}_L \end{bmatrix} = \begin{bmatrix} y_1^1 & y_1^2 & \cdots & y_1^N \\ \vdots & \vdots & & \vdots \\ y_l^1 & y_l^2 & \cdots & y_l^N \\ \vdots & \vdots & & \vdots \\ y_L^1 & y_L^2 & \cdots & y_L^N \end{bmatrix} \in \mathbb{R}^{L \times N}$$

forecasting horizon $h = 1$

$$\mathbf{y}_{L+1} = [y_{L+1}^1 y_{L+1}^2 \cdots y_{L+1}^N] \in \mathbb{R}^{1 \times N}$$



Compute a **forecasting model** $\mathcal{M}$ based on historical observations of the multiple time series $\mathbf{Y}$ to predict the future values y_{L+1}



Forecasting models

Statistical

- ✓ Block Hankel tensor-based autoregressive integrated moving average (BHT-ARIMA)

Machine learning

- ✓ Support vector regression (SVR)
- ✓ Random forest regression (RF)

Neural networks

- ✓ Long short-term memory (LSTM)
- ✓ Bidirectional long short-term memory (Bi-LSTM)
- ✓ Convolutional neural network (CNN)
- ✓ Echo state network (ESN)

Forecasting protocol

L + 1 time samples

N IoT sensors

Initial multiple time series dataset



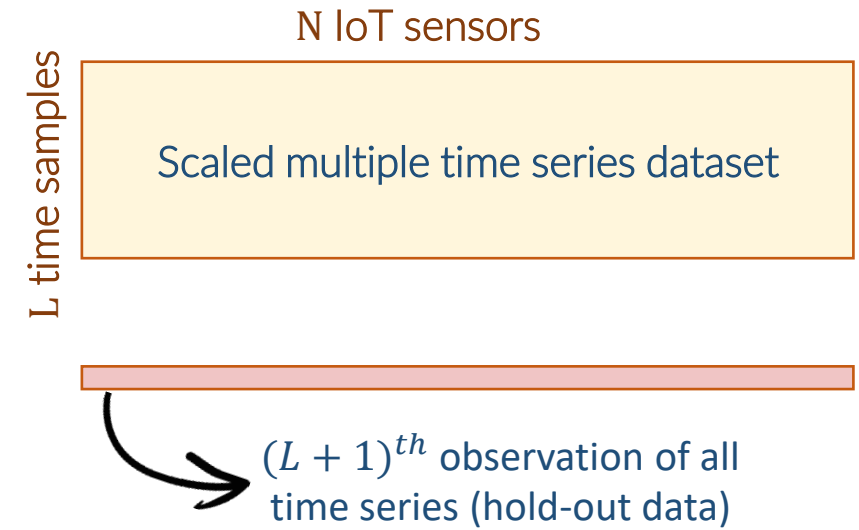
min-max scaling

$$\mathbf{y}_{sc}^n = \frac{\mathbf{y}^n - \mathbf{y}_{min}^n}{\mathbf{y}_{max}^n - \mathbf{y}_{min}^n}, \quad n = 1, \dots, N$$

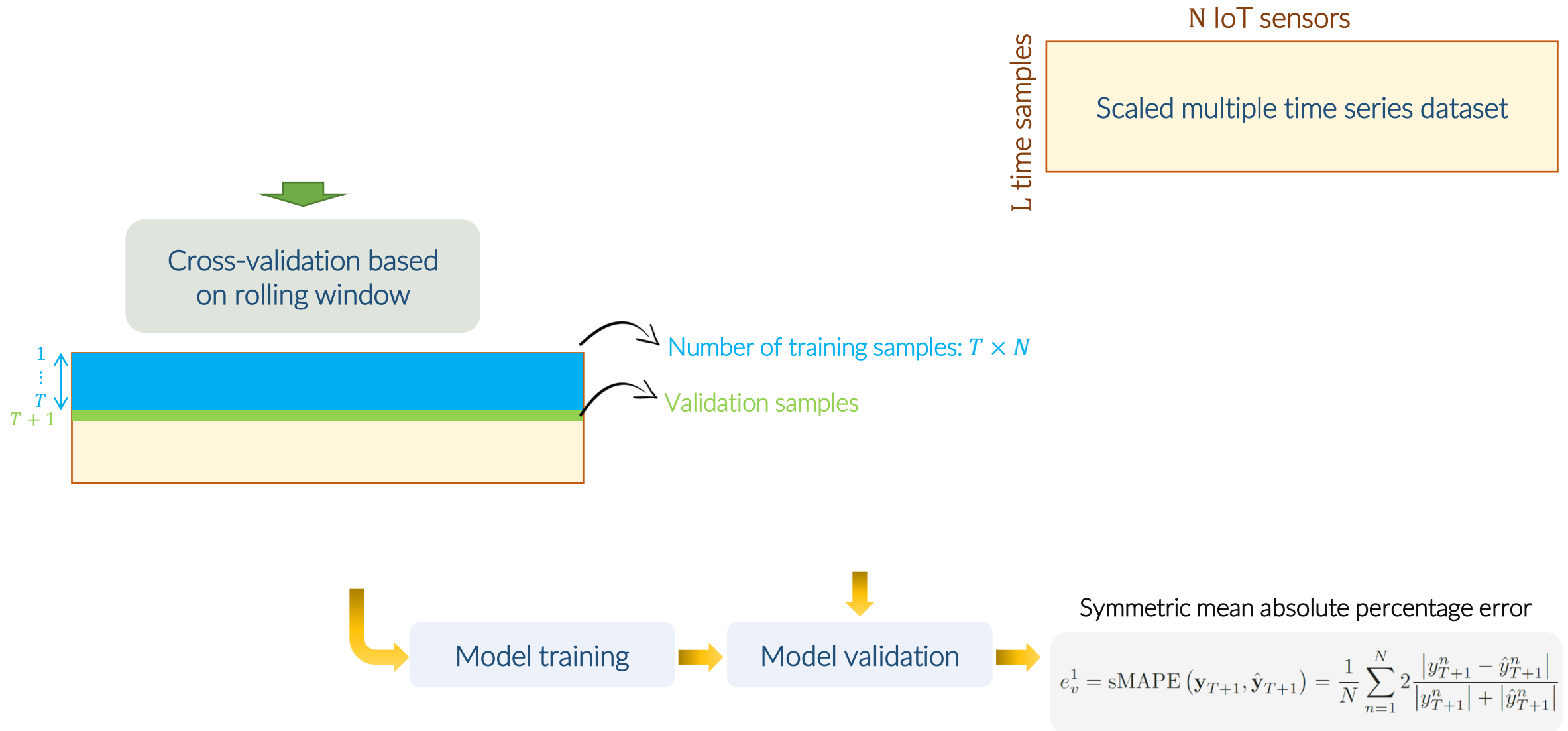


Scaled multiple time series dataset

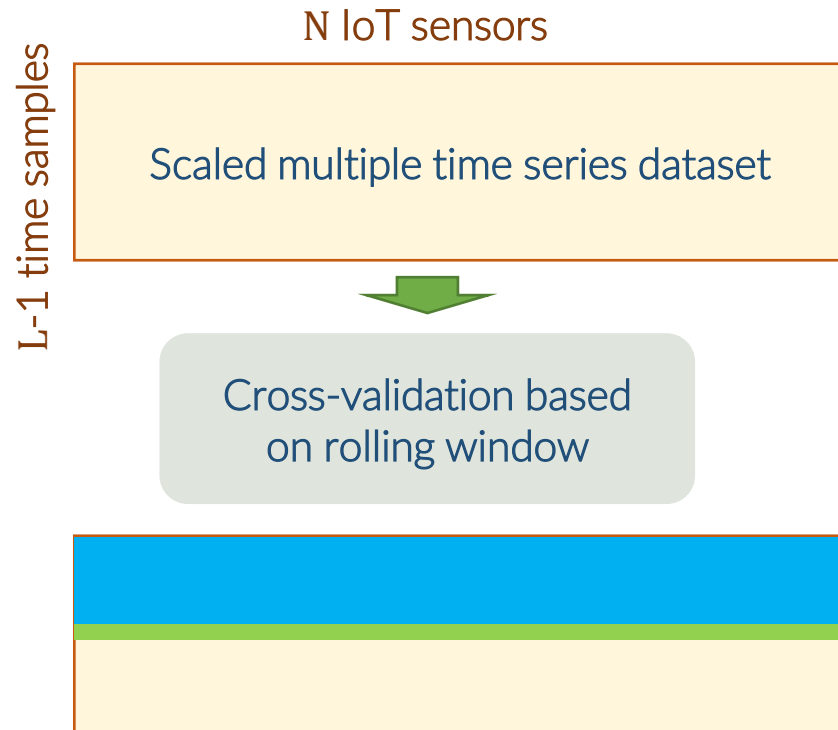
Forecasting protocol



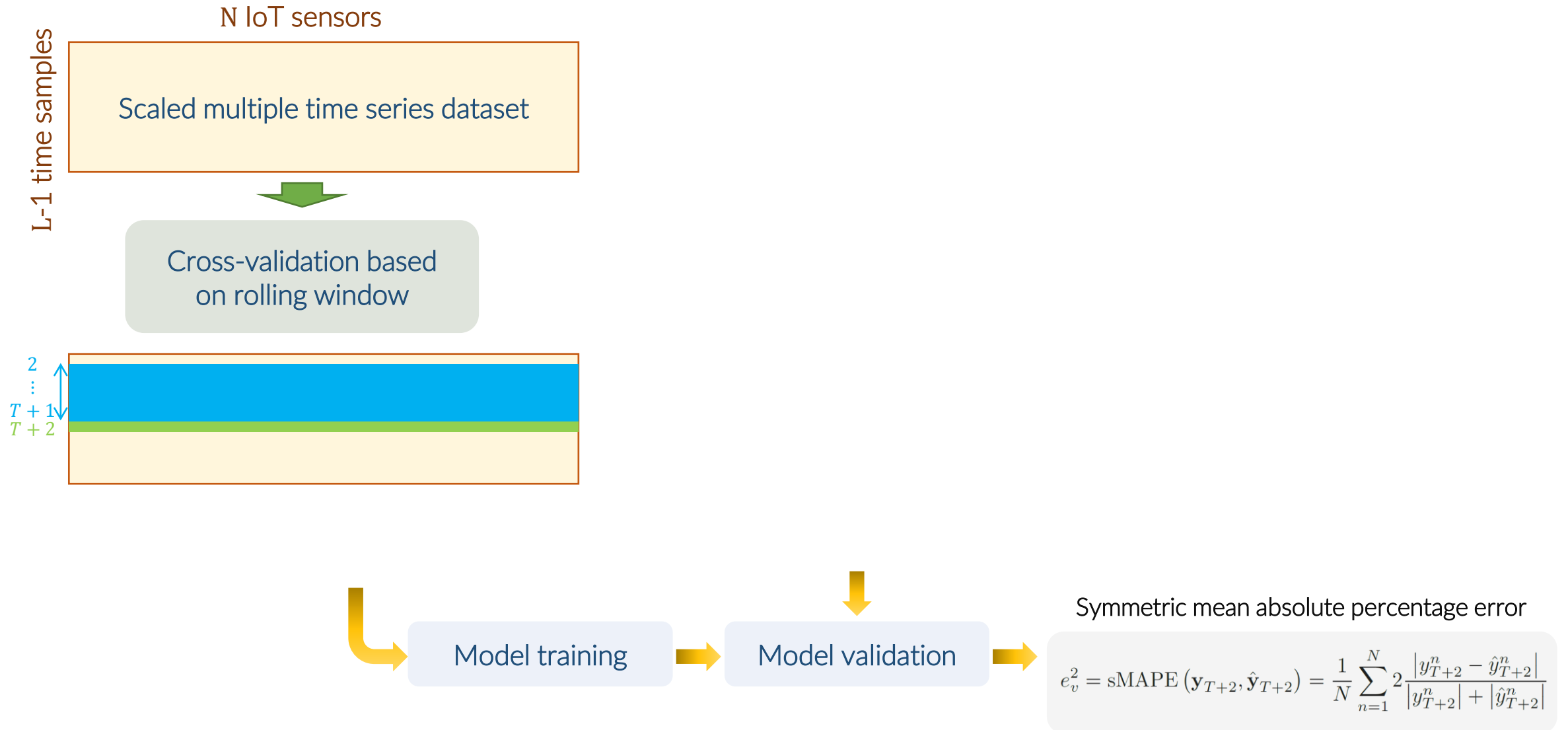
Forecasting protocol



Forecasting protocol



Forecasting protocol



Forecasting protocol

N IoT sensors

L-1 time samples

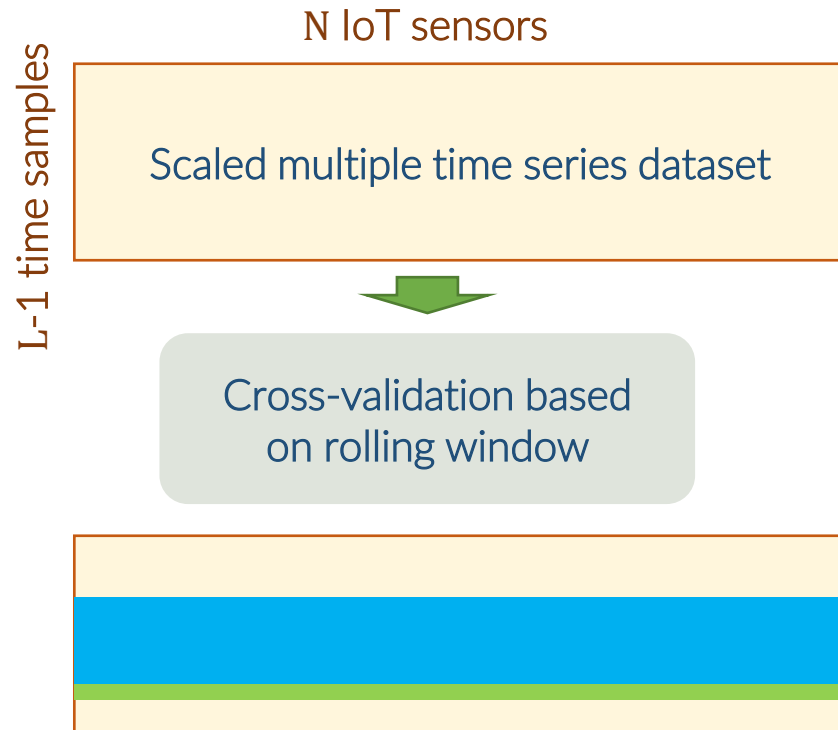
Scaled multiple time series dataset

Cross-validation based
on rolling window

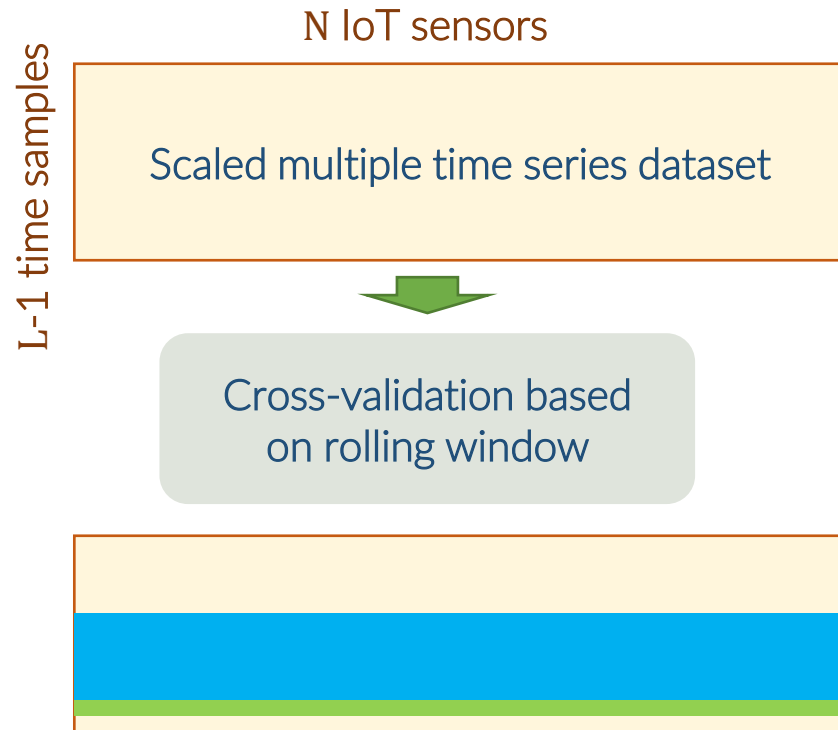
3
⋮
 $T+2$
 $T+3$



Forecasting protocol



Forecasting protocol



Forecasting protocol

N IoT sensors

L-1 time samples

Scaled multiple time series dataset

Cross-validation based
on rolling window

$L-T$
 $\vdots$
 $L-1$
 L

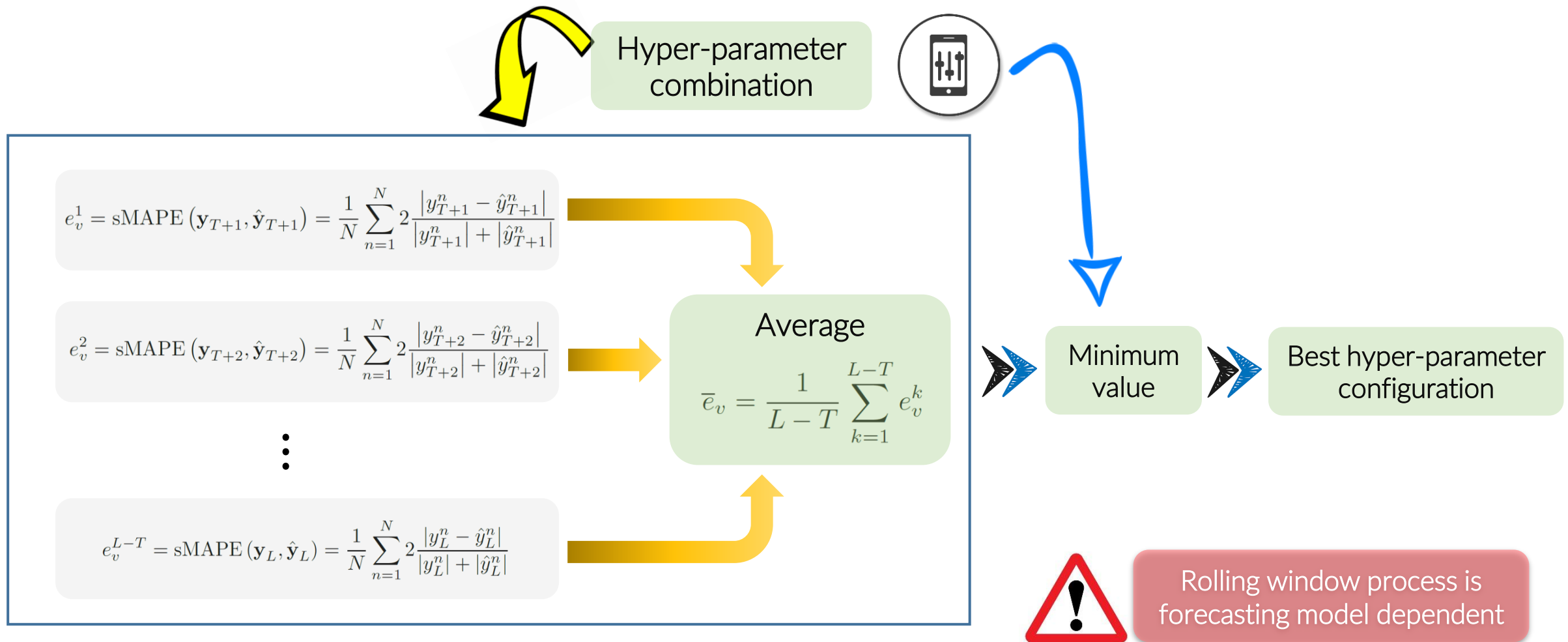
Model training

Model validation

Symmetric mean absolute percentage error

$$e_v^{L-T} = \text{sMAPE}(\mathbf{y}_L, \hat{\mathbf{y}}_L) = \frac{1}{N} \sum_{n=1}^N 2 \frac{|y_L^n - \hat{y}_L^n|}{|y_L^n| + |\hat{y}_L^n|}$$

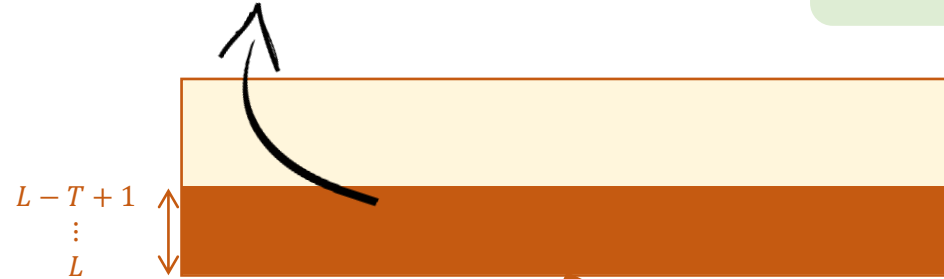
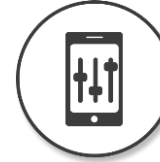
Forecasting protocol



Forecasting protocol

The model with the best hyper-parameter configuration is used to train the last T training data

Best hyper-parameter configuration



$\mathcal{M}$

Predict the next multiple time series sample $L + 1$

Prediction error

$(L + 1)^{th}$ observation of all time series (hold-out data)

$$\text{sMAPE}(\mathbf{y}_{L+1}, \hat{\mathbf{y}}_{L+1}) = \frac{1}{N} \sum_{n=1}^N 2 \frac{|y_{L+1}^n - \hat{y}_{L+1}^n|}{|y_{L+1}^n| + |\hat{y}_{L+1}^n|}$$

$$\text{MAAPE}(\mathbf{y}_{L+1}, \hat{\mathbf{y}}_{L+1}) = \frac{1}{N} \sum_{n=1}^N \arctan \left(\left| \frac{y_{L+1}^n - \hat{y}_{L+1}^n}{y_{L+1}^n} \right| \right)$$

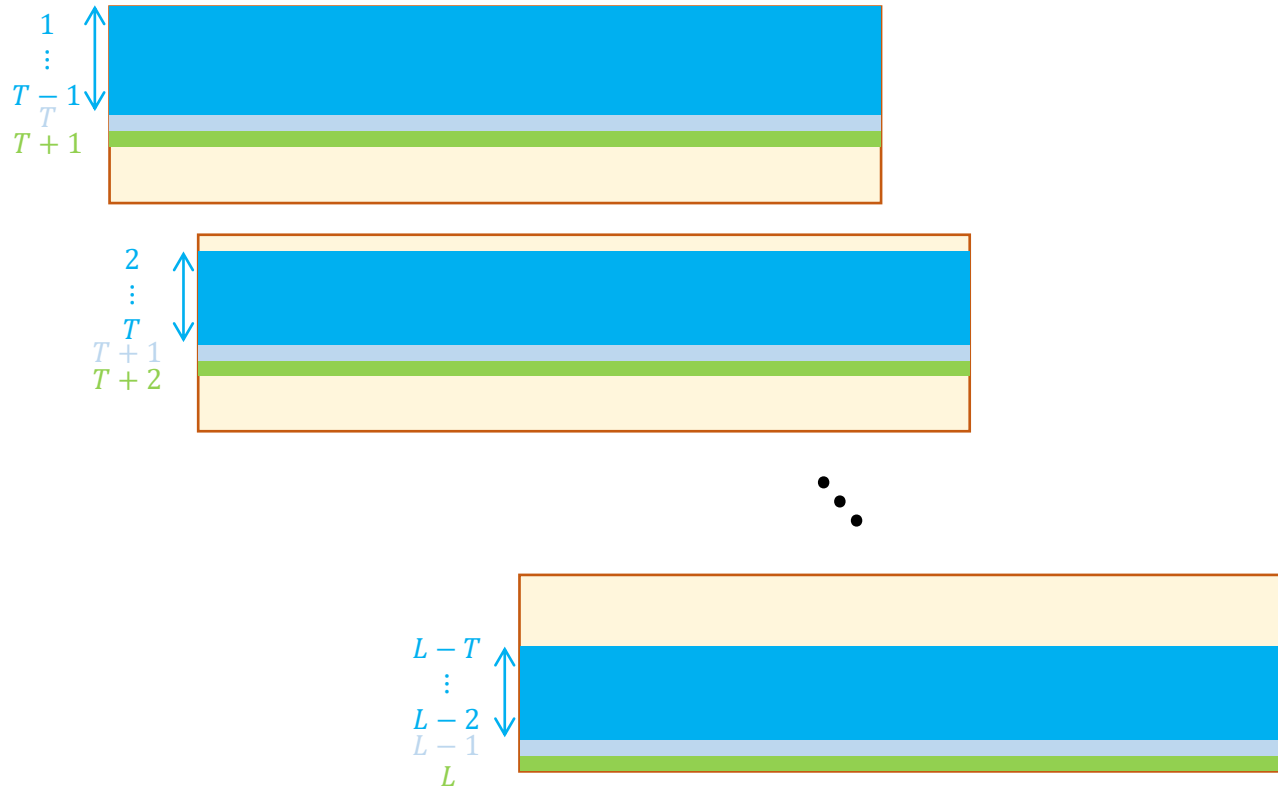
$$\text{MASE}(\mathbf{y}_{L+1}, \hat{\mathbf{y}}_{L+1}) = \frac{1}{N} \sum_{n=1}^N \frac{|y_{L+1}^n - \hat{y}_{L+1}^n|}{\frac{1}{L-1} \sum_{l=2}^L |y_l^n - \hat{y}_l^n|}$$

Forecasting protocol

Rolling window: matrix case



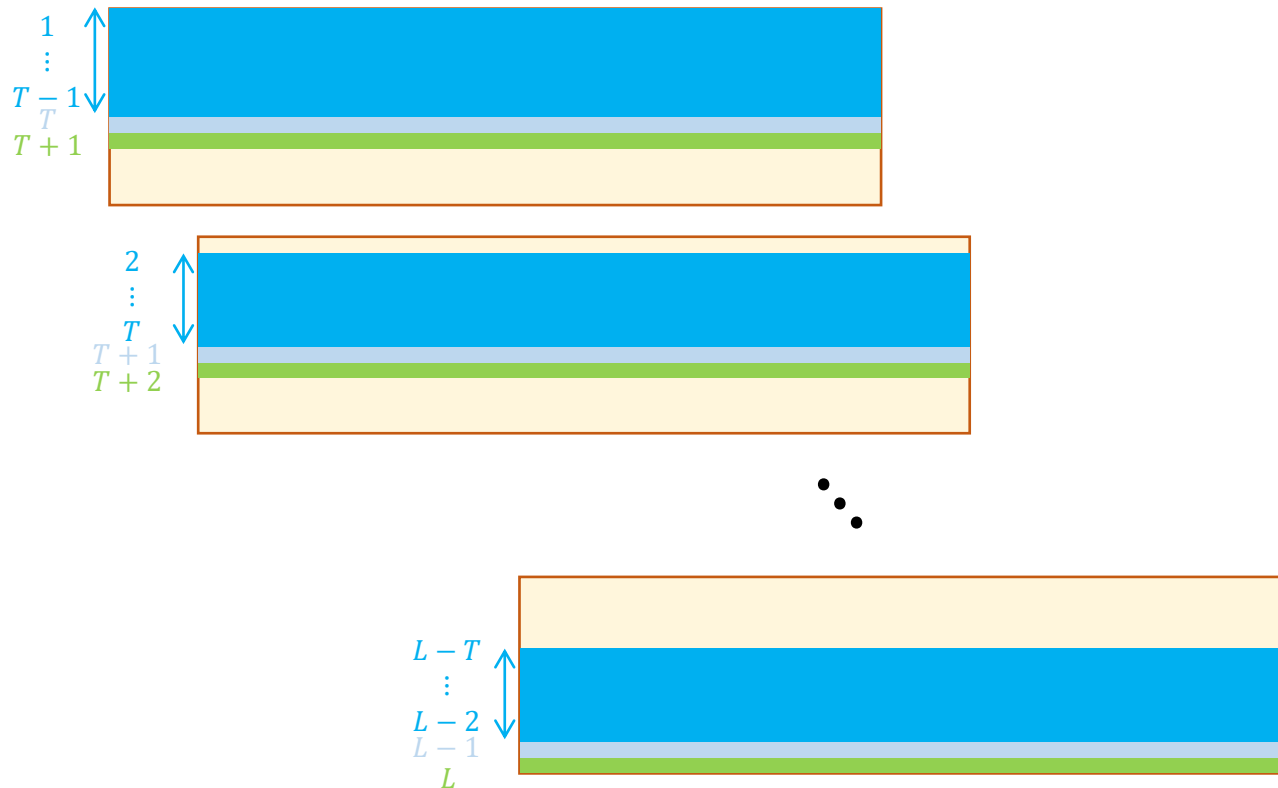
SVR & RF



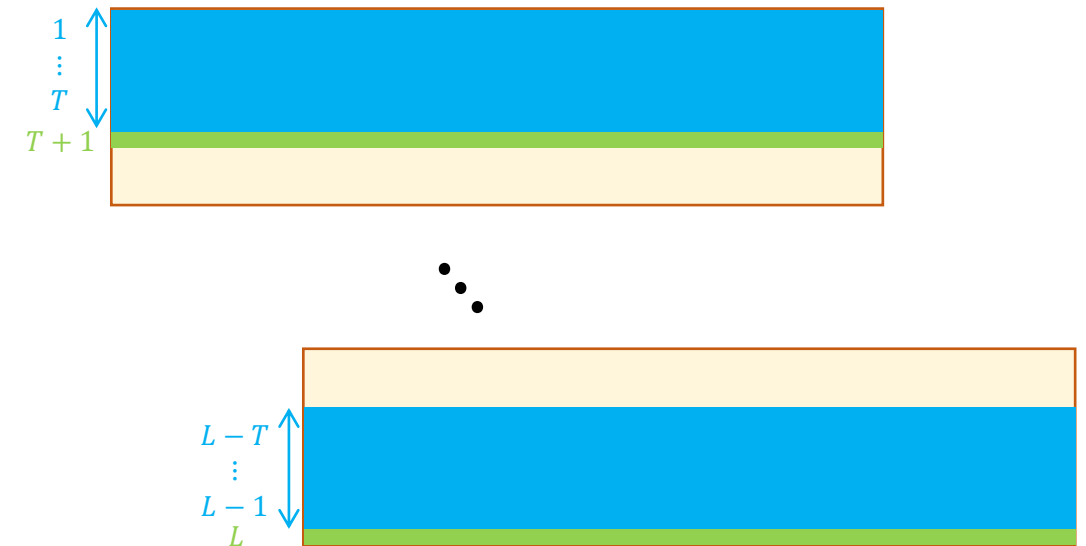
Forecasting protocol

Rolling window: matrix case

 SVR & RF

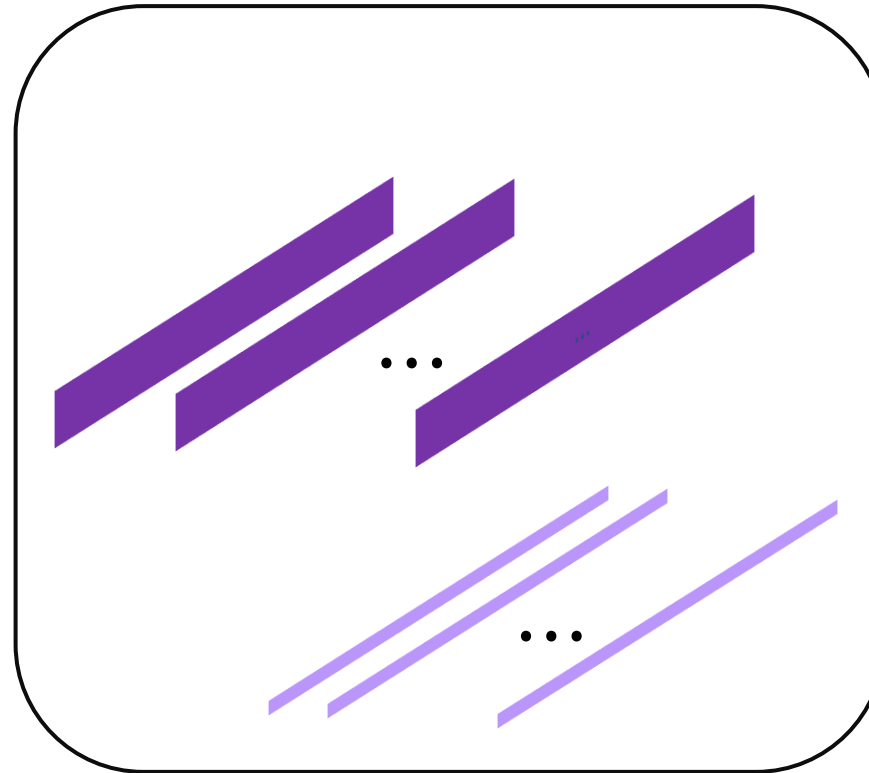
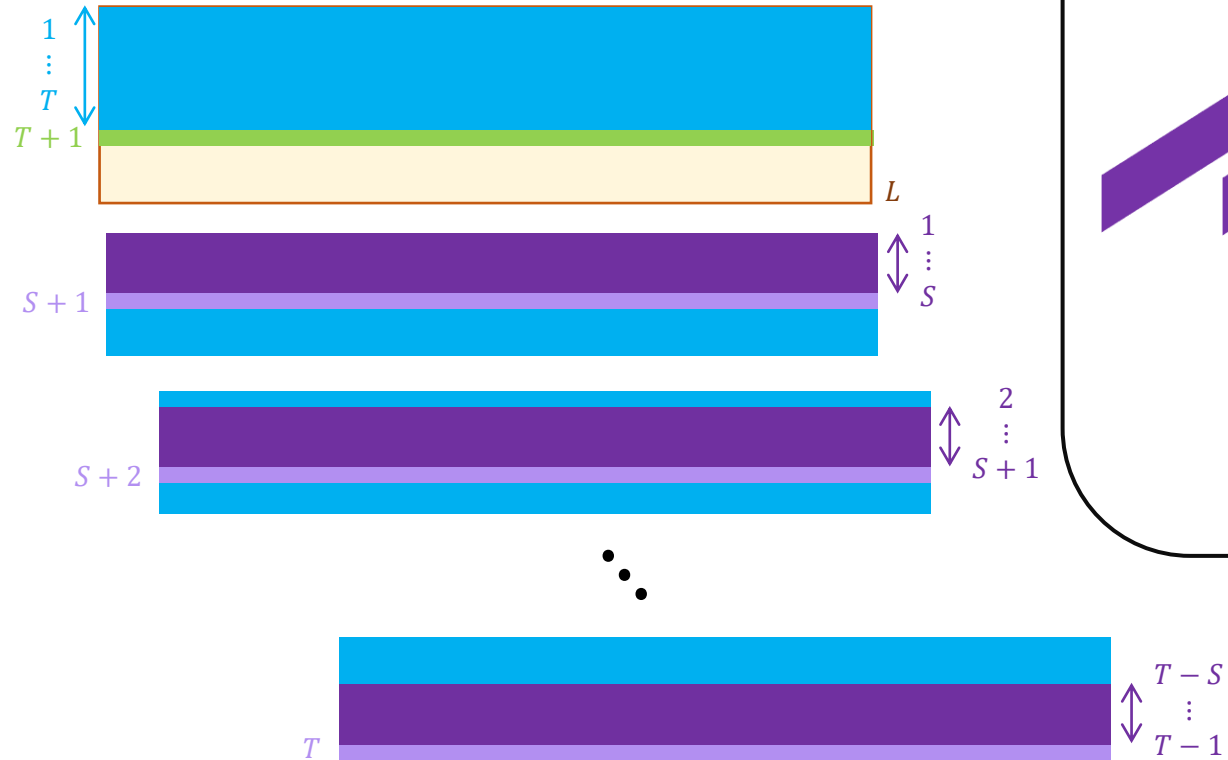


 BHT-ARIMA





LSTM, Bi-LSTM & CNN



Model training

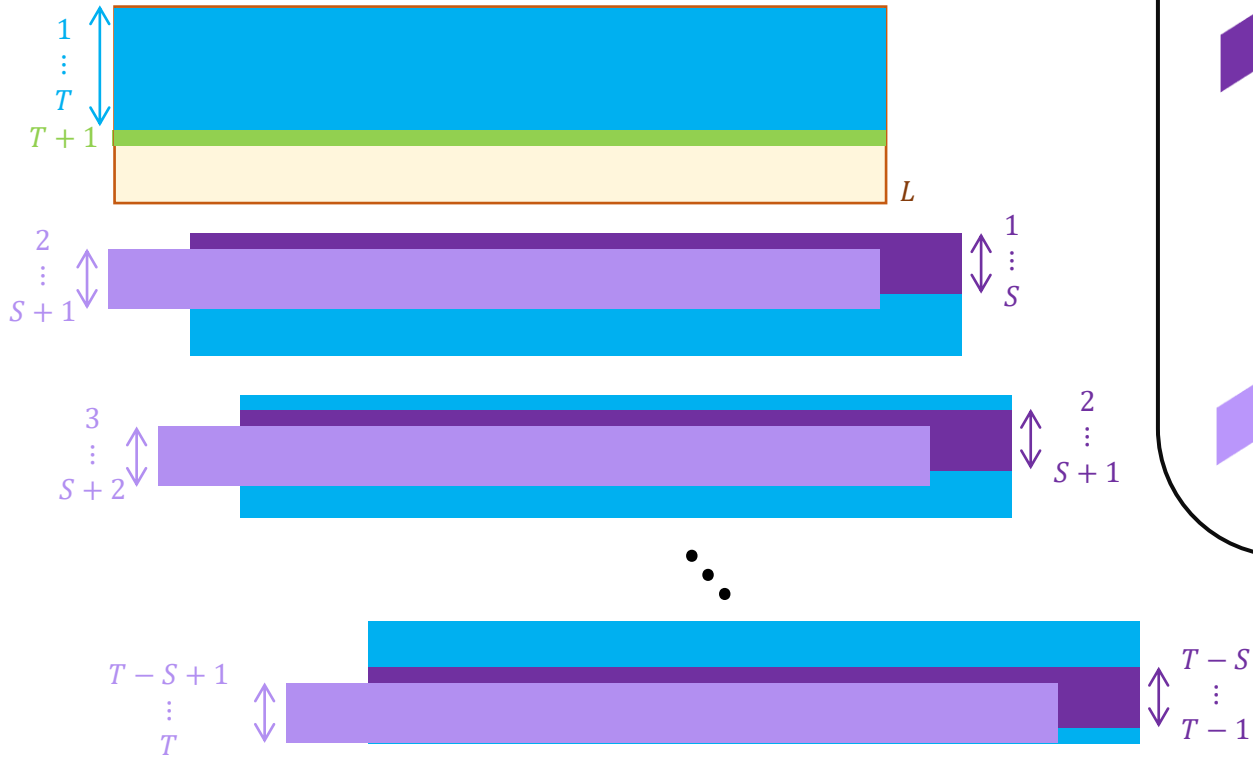
Model validation

Forecasting protocol

Rolling window: *Matrix List Case*



ESN



Model training

Model validation

Energy
Consumption
Fraunhofer
D1

Number of time series	314
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Number of time samples per series	365
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Time resolution	daily
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Electricity Load
Diagrams
D2

Number of time series	320
--------------------------	-----

Number of time samples per series	1096
---	------

Time resolution	daily
--------------------	-------

Guangzhou
Traffic
D3

Number of time series	206
--------------------------	-----

Number of time samples per series	1464
---	------

Time resolution	hourly
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San Francisco
Traffic
D4

Number of time series	862
--------------------------	-----

Number of time samples per series	104
---	-----

Time resolution	weekly
--------------------	--------

London Smart
Meters
D5

Number of time series	504
--------------------------	-----

Number of time samples per series	9983
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Time resolution	half hourly
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- Train/validation split: 80/20
- Apply a model-dependent rolling window process
 - ✓ Hyper-parameter grid search is performed for each model per dataset
- Number of multiple time series samples: 40 & 90
- 15 Monte Carlo runs
 - ✓ During each iteration examine a different part of the multiple time series dataset
 - ✓ Prediction error averaged over all Monte Carlo runs



Check EUSIPCO
paper

Experimental results

DATASET		D1		D2		D3		D4		D5	
# Sim. data		40	90	40	90	40	90	40	90	40	90
BHT-ARIMA	sMAPE	0.3369	0.3554	0.0969	0.0819	0.0797	0.1102	0.0742	0.1022	0.5209	0.4683
	MAAPE	0.2888	0.3035	0.0986	0.0756	0.0765	0.1050	0.0724	0.1025	0.4298	0.3988
	MASE	0.1806	0.1905	0.0503	0.0247	0.3169	0.3569	0.1877	0.2576	0.7366	0.3829
SVR	sMAPE	0.3683	0.3816	0.0931	0.0765	0.0655	0.0930	0.0795	0.0946	0.4705	0.5047
	MAAPE	0.3275	0.3424	0.0933	0.0732	0.0666	0.0935	0.0766	0.0952	0.4383	0.4898
	MASE	0.1867	0.1945	0.0421	0.0250	0.2697	0.3190	0.1978	0.2267	0.3878	0.4271
RF	sMAPE	0.3200	0.3417	0.1026	0.0813	0.0601	0.0921	0.0814	0.1076	0.4649	0.4353
	MAAPE	0.2907	0.3086	0.1033	0.0782	0.0615	0.0923	0.0788	0.1068	0.4461	0.4281
	MASE	0.1892	0.1927	0.0459	0.0248	0.2324	0.2978	0.2037	0.2617	0.3935	0.3673
LSTM	sMAPE	0.5551	0.6310	0.1668	0.2151	0.4762	0.5625	0.1771	0.2652	0.9009	0.8736
	MAAPE	0.3672	0.4038	0.1373	0.1597	0.3281	0.3675	0.1488	0.2204	0.4959	0.4842
	MASE	0.3111	0.3557	0.0849	0.0885	1.8338	2.0560	0.4068	0.5719	0.6320	0.6127
Bi-LSTM	sMAPE	0.5557	0.6299	0.1667	0.2138	0.4761	0.5515	0.1770	0.3246	0.9017	0.8734
	MAAPE	0.3682	0.4048	0.1373	0.1590	0.3280	0.3623	0.1487	0.2577	0.4963	0.4843
	MASE	0.3112	0.3558	0.0849	0.0883	1.8338	2.0253	0.4067	0.7376	0.6321	0.6124
CNN	sMAPE	0.5555	0.6314	0.1668	0.2150	0.4738	0.5619	0.1771	0.2639	0.9010	0.8732
	MAAPE	0.3680	0.4046	0.1374	0.1596	0.3266	0.3672	0.1488	0.2191	0.4958	0.4845
	MASE	0.3113	0.3562	0.0849	0.0883	1.8235	2.0542	0.4069	0.5678	0.6322	0.6124
ESN	sMAPE	0.3510	0.4262	0.1021	0.0749	0.0588	0.0825	0.0825	0.1216	0.5154	0.6125
	MAAPE	0.3084	0.3657	0.1030	0.0719	0.0601	0.0830	0.0801	0.1206	0.4562	0.5251
	MASE	0.1897	0.2240	0.0474	0.0256	0.2322	0.2954	0.2054	0.2714	0.4239	0.5068

Take-home messages

-  Short-term forecasting of multiple time series in IoT environments
 - ✓ #time-series >> #observations
 - ✓ Forecasting protocol especially designed for multiple time series at the IoT edge
-  Seven forecasting models are examined
 - ✓ Statistical, machine learning and neural networks
-  Five real-world IoT datasets
-  Statistical & machine learning methods in general perform better VS typical neural network architectures



<https://github.com/pcharala/multiple-timeseries-forecasting>